ZIP models

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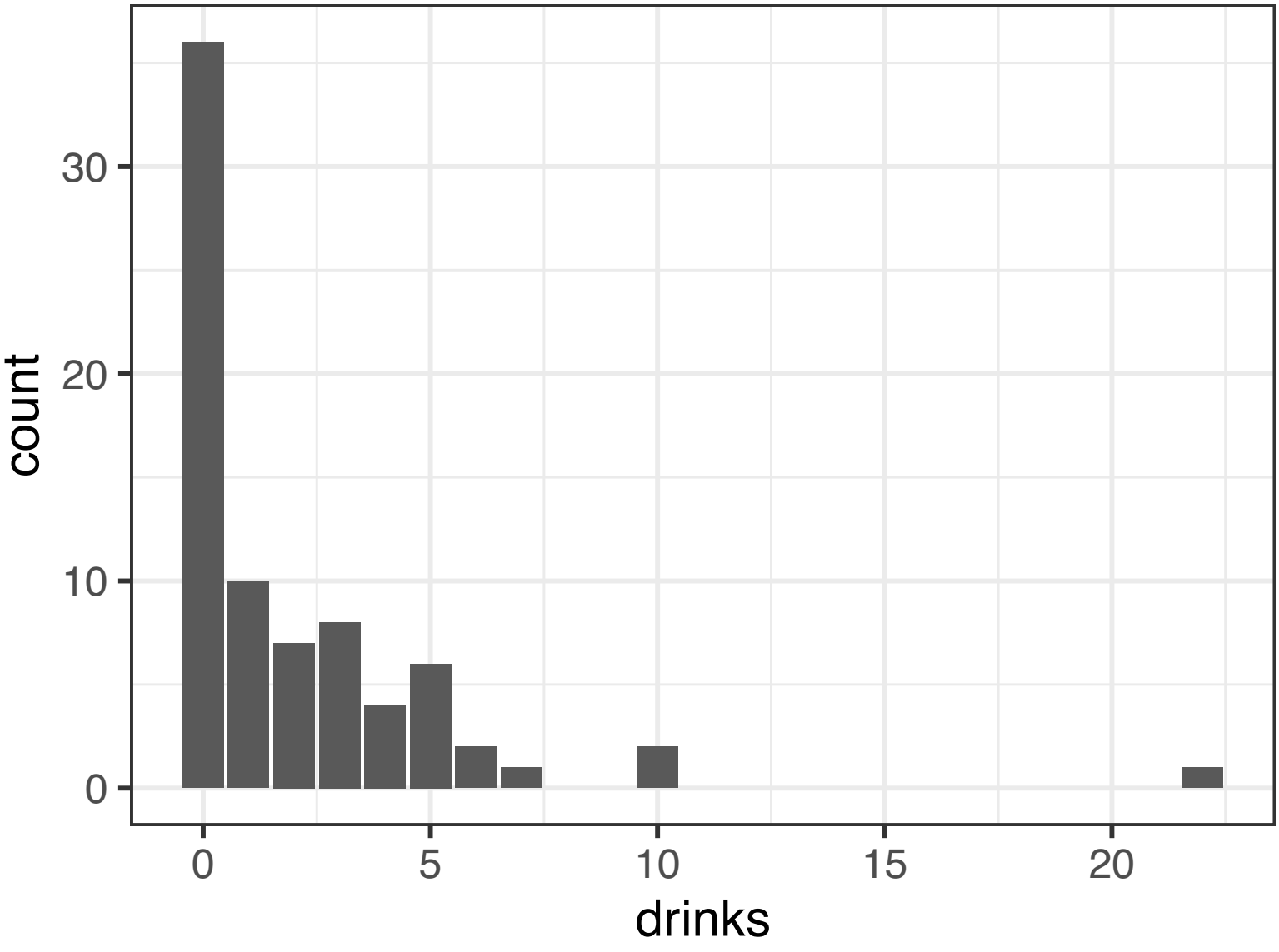
Last time: weekend drinking

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- `drinks`: number of drinks the student reports consuming
- `sex`: whether the student identifies as male
- `OffCampus`: whether the student lives off campus
- `FirstYear`: whether the student is a first-year student

Our goal: model the number of drinks students report consuming.

EDA: drinks



Zero-inflated Poisson (ZIP) model

random component:

$$P(Y_i = y \mid x_i) = \begin{cases} p_i + (1-p_i)e^{-\lambda_i} & y = 0 \\ (1-p_i)\dfrac{e^{-\lambda_i}\lambda_i^{y}}{y!} & y > 0 \end{cases}$$

Systematic component:

$$\log(\lambda_i) = x_i^T \beta$$

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^T \gamma$$

In R

```
library(pscl)

m1 <- zeroinfl(drinks ~ sex + FirstYear + OffCampus,
               dist = "poisson", data = wdrinks)

m1$coefficients
```

```
## $count
##   (Intercept)           sexm FirstYearTRUE OffCampusTRUE
##     0.8010435      0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)           sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078     0.89197114   -1.69137146
```

Activity

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated probability that a female first-year student who lives on campus never drinks? $\leftarrow \quad Z_i = 1$

$$\hat{p}_i = \hat{P}(Z_i = 1 \mid x_i)$$

$$= \frac{\exp\{-0.396 + 0.892\}}{1 + \exp\{-0.396 + 0.892\}} \approx 0.623$$

Activity

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated average number of drinks for a second-year male student who lives off campus and sometimes drinks?

want $E[Y_i \mid Z_i = 0, x_i] = \lambda_i$ $\quad Z_i = 0$

$$\hat{\lambda}_i = \exp\{0.801 + 0.983 + 0.372\} \approx 8.64$$

Activity

$1 - (0.22 + (1-0.22)e^{-7.34}) \approx 0.779$

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated probability that a male first-year student who lives off campus had at least one drink last weekend?

$P(Y_i > 0 \mid x_i) = 1 - P(Y_i = 0 \mid x_i)$

$P(Y_i = 0 \mid x_i) = p_i + (1-p_i)e^{-\lambda_i}$

$\hat{\lambda}_i = \exp\{0.801 + 0.983 - 0.162 + 0.372\} \approx 7.34$

$\hat{p}_i = \frac{\exp\{-0.396 - 0.071 + 0.892 - 1.691\}}{1 + \exp\{\dots\}} \approx 0.22$

Activity: fitting ZIP models

Work on the class activity (course website).